

LEEK THORNCLIFFE, United Kingdom

WMO: 033300

Lat: 53.1278N

Lon: 1.9814W

Elev: 981

StdP: 14.18

Time Zone: 0.00 (EUW)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	27.0	20.9	16.2	28.0	23.1	18.0	29.1	31.2	41.7	28.6	40.0	9.2	30	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	74.6	64.0	70.9	62.1	68.0	60.8	65.4	71.8	63.5	68.9	61.8	66.4	9.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	88.8	67.9	61.2	83.8	65.8	59.8	79.7	63.9	30.7	71.9	29.3	69.0	28.1	66.4	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.0	25.0	22.3	DB	21.7	80.0	3.1	3.7	19.5	82.6	17.7	84.7	16.0	86.8	13.8	89.4
			WB	20.9	69.1	2.9	2.1	18.7	70.6	17.0	71.8	15.4	73.0	13.3	74.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	37.7	37.9	40.9	45.5	50.5	55.8	59.3	57.9	54.6	49.3	42.3	39.1
	DBStd	9.23	5.12	5.55	5.51	5.34	5.47	4.81	4.92	3.81	4.39	4.81	5.35	5.81
	HDD50	1874	382	338	286	154	59	5	0	0	9	68	234	339
	HDD65	6365	847	758	748	585	450	278	191	223	315	486	680	804
	CDD50	1007	0	1	3	19	74	180	288	244	146	47	4	1
	CDD65	20	0	0	0	0	0	3	14	2	1	0	0	0
	CDH74	120	0	0	0	0	1	23	83	11	2	0	0	0
	CDH80	7	0	0	0	0	0	0	7	0	0	0	0	0
Wind	WSAvg	10.6	11.8	11.3	11.7	11.5	10.8	9.6	8.9	8.8	9.5	11.0	10.8	11.1
Precipitation	PrecAvg	40.6	3.8	3.2	2.6	2.8	2.8	3.1	3.1	3.3	3.1	4.3	4.1	4.4
	PrecMax	54.1	7.7	7.6	5.7	6.5	5.2	7.8	7.5	9.5	6.2	9.4	8.7	7.8
	PrecMin	30.6	0.3	0.7	0.4	0.4	0.6	0.8	1.3	0.8	0.6	1.4	1.8	1.1
	PrecStd	6.2	1.8	1.5	1.3	1.5	1.3	1.7	1.3	1.7	1.4	1.9	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	55.0	58.5	66.8	72.6	77.3	80.1	75.3	71.3	64.2	56.9	53.4
		MCWB	51.0	48.8	51.0	55.5	61.6	64.2	66.0	64.5	63.8	59.4	55.7	51.8
	2%	DB	49.6	49.7	54.0	60.5	67.8	71.8	76.2	69.6	66.7	60.2	54.1	51.3
		MCWB	48.5	46.8	48.7	52.2	59.8	62.1	64.8	62.1	60.9	57.8	53.0	50.3
	5%	DB	47.4	47.6	51.0	56.8	63.5	68.3	72.4	67.0	63.9	58.0	52.1	49.5
		MCWB	46.3	45.9	47.9	50.4	56.4	60.4	63.2	60.9	59.0	56.2	51.1	48.6
	10%	DB	45.5	45.7	48.6	54.0	59.6	65.0	68.7	64.6	61.6	56.1	50.0	47.6
		MCWB	44.6	44.2	46.0	49.0	53.6	58.1	61.3	59.4	57.5	54.4	49.1	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	49.5	52.1	57.2	63.2	67.2	68.4	66.2	64.7	60.6	55.8	52.1
		MCDB	51.7	51.3	55.5	64.7	70.0	74.0	76.8	71.9	69.0	63.0	56.8	52.6
	2%	WB	48.8	47.9	50.0	53.5	59.9	63.3	65.9	63.9	62.2	58.4	53.3	50.6
		MCDB	49.4	49.5	52.8	58.9	66.9	69.2	73.7	68.0	65.6	60.0	53.9	51.2
	5%	WB	46.6	46.2	48.3	51.6	57.2	61.0	63.9	62.1	60.0	56.6	51.3	48.8
		MCDB	47.1	47.4	50.7	55.7	62.2	66.8	70.4	65.5	62.6	57.7	51.9	49.4
	10%	WB	44.7	44.4	46.5	49.6	54.9	59.1	62.1	60.5	58.4	54.7	49.2	46.7
		MCDB	45.3	45.5	48.3	53.7	58.6	63.7	67.4	63.3	60.6	55.7	49.8	47.4
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	10.5	12.7	13.5	13.7	13.3	11.8	11.4	9.3	8.2	7.3
		MCDBR	9.8	9.6	12.6	16.0	18.0	18.6	19.5	14.7	13.9	9.8	9.6	9.2
		MCWBR	10.0	8.1	8.7	10.3	11.7	11.2	10.6	8.6	8.8	7.8	9.1	8.9
	5% WB	MCDBR	9.8	9.0	11.8	14.6	16.3	16.9	17.6	12.9	12.6	9.5	9.6	9.2
		MCWBR	10.1	8.1	8.9	10.4	11.4	11.2	10.5	8.6	9.3	8.3	9.4	9.1
Clear-Sky Solar Irradiance	taub	0.313	0.328	0.357	0.382	0.385	0.384	0.384	0.370	0.355	0.337	0.314	0.301	
	taud	2.351	2.401	2.321	2.263	2.292	2.316	2.317	2.376	2.424	2.450	2.410	2.323	
	Ebn at Noon	210	247	261	269	274	275	272	270	260	240	210	190	
	Edh at Noon	19	24	32	38	39	38	38	34	29	23	17	16	
All-Sky Solar Radiation	RadAvg	216	431	782	1207	1515	1590	1521	1250	919	521	264	170	
	RadStd	20	49	84	130	101	154	158	85	77	45	29	17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page